EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.
DATE:26/03/2025	

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their **username** and **password** in a **login form**.
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

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2. Steps for Implementing Session Tracking with HttpSession:

1. **Create the Login Form (HTML):**
    - Create an HTML form with two input fields for **username** and **password**.
    - Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
    - Write a Java servlet class to handle **login** requests.
    - The servlet will:
        - Retrieve the **username** and **password** from the form.
        - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
        - Create a new **HTTP session** for the user if the credentials are valid.
        - Set session attributes such as the user's **username** to track the session.

3. **Validate User Credentials:**
    - In the servlet, check if the **username** and **password** match the expected values.
    - If valid, create a session and store the **username** in the session.
    - If invalid, redirect the user back to the login page with an error message.
4. **Create and Track User Session:**
    - Upon successful login, use the HttpSession object to store the session data.
    - Use the session.setAttribute("username", username) method to save the **username**.
    - Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.
5. **Redirect to a Protected Page:**
    - After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
    - The protected page should check if the user is authenticated by checking the **session attribute**.
6. **Logout and Session Invalidation:**
    - Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
    - After logout, redirect the user back to the login page.
7. **Session Expiration:**
    - If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Program :

Index.html :

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Login Page</title>
```

```html
<style>
body{
    margin:0;
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    font-family:Arial, sans-serif;
    background: linear-gradient(135deg,#667eea,#764ba2);
}
.container{
    background:white;
    padding:40px;
    width:350px;
    border-radius:15px;
    box-shadow:0 15px 30px rgba(0,0,0,0.2);
    text-align:center;
}
h2{
    margin-bottom:20px;
    color:#333;
}
input{
    width:90%;
    padding:12px;
    margin:10px 0;
    border-radius:8px;
    border:1px solid #ccc;
}
button{
    width:95%;
    padding:12px;
    border:none;
    border-radius:8px;
    background:#667eea;
    color:white;
    font-size:16px;
    cursor:pointer;
}
button:hover{
    background:#5641c5;
}
</style>
</head>
<body>
<div class="container">
<h2>🔐 Login</h2>
<form action="LoginServlet" method="post">
<input type="text" name="username" placeholder="Enter Username" required>
<input type="password" name="password" placeholder="Enter Password" required>
<button type="submit">Login</button>
</form>
</div>
</body>
</html>
```

Dashboard.jsp:

```jsp
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="javax.servlet.http.*" %>

<%

    HttpSession sessionObj = request.getSession(false);

    if(sessionObj == null || sessionObj.getAttribute("username") == null){
        response.sendRedirect("login.html");
        return;
    }

    String user = (String) sessionObj.getAttribute("username");
%>

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Dashboard</title>

<style>
body{
    margin:0;
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    font-family:Arial, sans-serif;
    background: linear-gradient(135deg,#43cea2,#185a9d);
}

.box{
    background:white;
    padding:40px;
    width:400px;
    border-radius:15px;
    text-align:center;
    box-shadow:0 15px 30px rgba(0,0,0,0.3);
}

h2{
    color:#185a9d;
}

p{
    font-size:16px;
    color:#333;
}

button{
    margin-top:20px;
    padding:12px 25px;
    border:none;
    border-radius:8px;
    background:#185a9d;
```

```
    color:white;
    font-size:16px;
    cursor:pointer;
    transition:0.3s;
}

button:hover{
    background:#0f3e6b;
}
</style>
</head>

<body>

<div class="box">
    <h2>Welcome <%= user %></h2>
    <p>You are successfully logged in.</p>

    <form action="LogoutServlet" method="post">
        <button type="submit">Logout</button>
    </form>
</div>

</body>
</html>
```

Web.xml:

```
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         version="3.1">

</web-app>
```

Login.html:

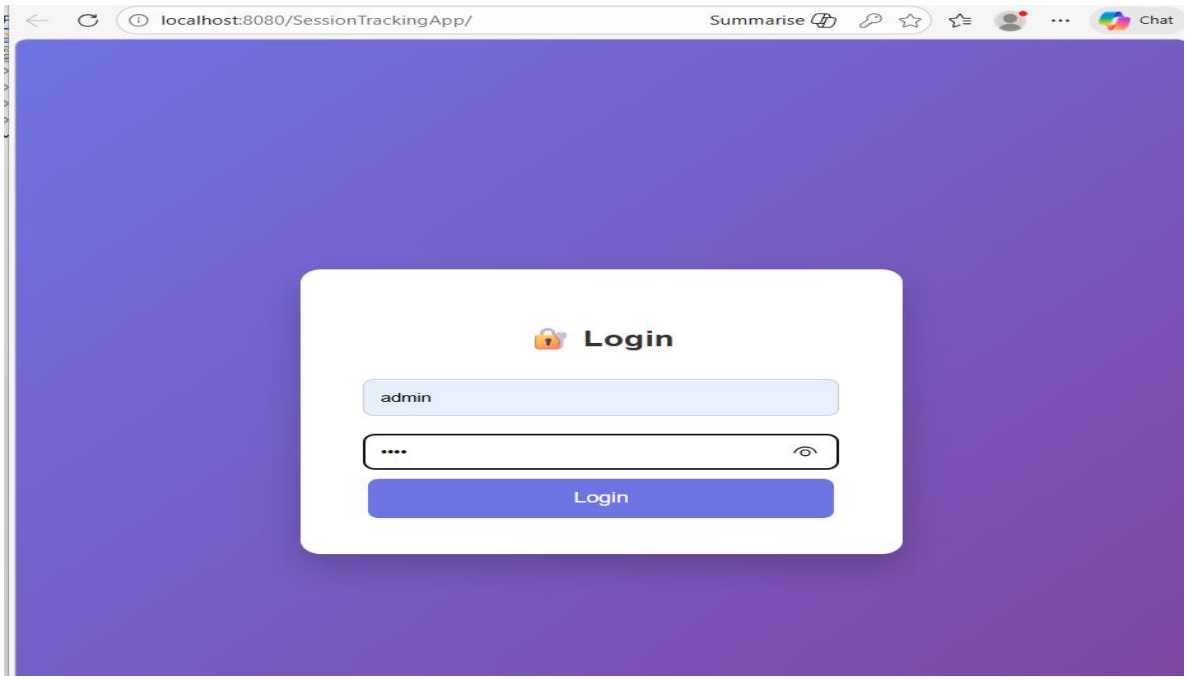
```
package com.hello;   // your package name (check yours)

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if(username.equals("admin") && password.equals("1234")) {

            HttpSession session = request.getSession();
```

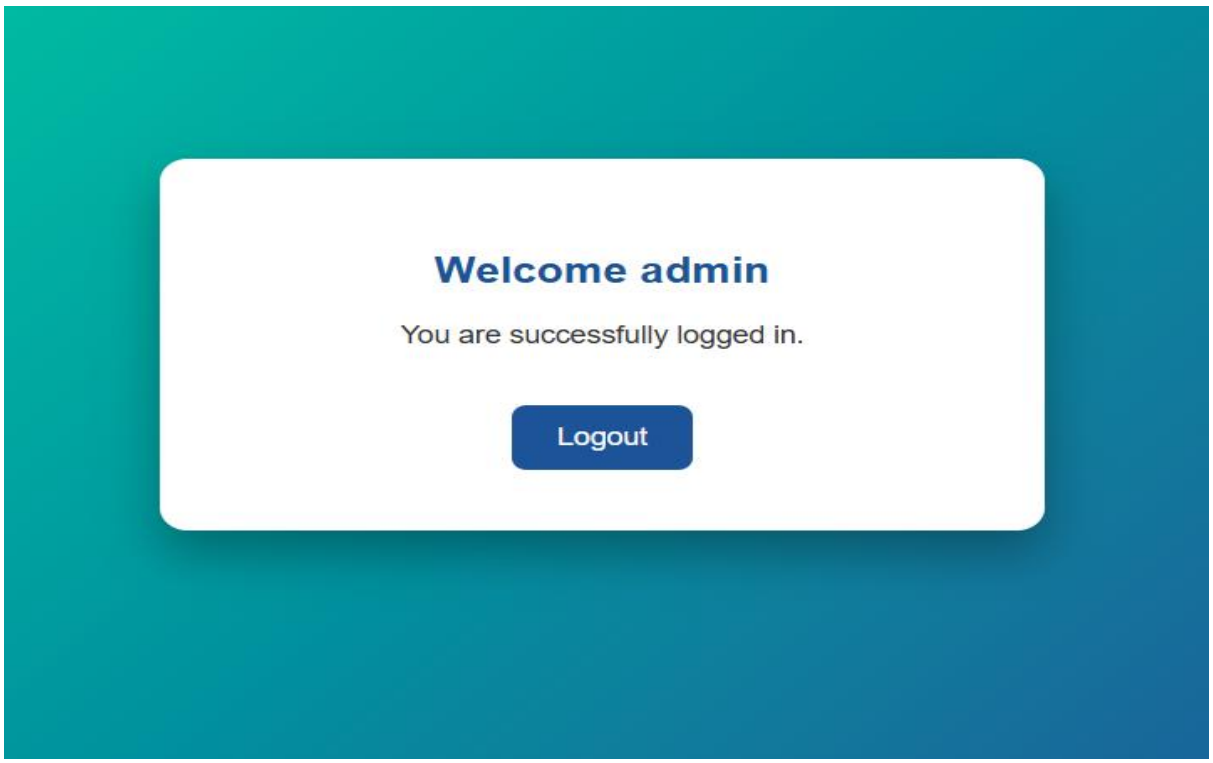
```
            session.setAttribute("username", username);

            response.sendRedirect("dashboard.jsp");

        } else {
            response.getWriter().println("Invalid Login");
        }
    }
}
```

Logout.html:

```
package com.hello;
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if(session != null){
            session.invalidate();
        }

        response.sendRedirect("login.html");
    }
}
```

Output:





RESULT:

A simple login system is created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.